



Cambridge International AS & A Level

CANDIDATE
NAME
CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

COMPUTER SCIENCE**9618/33**

Paper 3 Advanced Theory

May/June 2024**1 hour 30 minutes**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **12** pages. Any blank pages are indicated.



1 Real numbers are stored in a computer system using floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the denary value of the given normalised floating-point number.

Show your working.

Mantissa

0	1	0	0	1	1	1	1	0	0
---	---	---	---	---	---	---	---	---	---

Exponent

0	0	1	0	0	1
---	---	---	---	---	---

Working

.....

.....

.....

.....

Answer

[3]

(b) Calculate the normalised floating-point representation of -102.75 in this system.

Show your working.

Mantissa

--	--	--	--	--	--	--	--	--	--

Exponent

--	--	--	--	--	--

Working

.....

.....

.....

.....

.....

[3]





2 The TCP/IP protocol suite has four layers:

Transport, Application, Link, Internet

(a) Complete the diagram to show the correct order for these layers.

[2]

(b) Describe the function of the Transport layer.

.....

.....

.....

..... [2]

(c) Outline **one** protocol that is associated with the Application layer.

.....

.....

.....

..... [2]

3 (a) Explain what is meant by **non-composite** and **composite** data types.

.....

.....

.....

.....

.....

..... [3]





(b) Write **pseudocode** statements to declare the record data type `FootballClub` to hold data about football clubs in a league, to include:

- name of team
- date team joined the league
- main telephone number
- name of the manager
- number of members
- current position in the league.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

4 (a) Describe the sequential method of file **access**.

.....

.....

.....

..... [2]

(b) Explain how the sequential method of file access is applied to files with serial organisation and to files with sequential organisation.

.....

.....

.....

.....

.....

..... [3]



DO NOT WRITE IN THIS MARGIN



5 (a) Write this Reverse Polish Notation (RPN) in infix form:

$$5 \ 2 \ + \ 9 \ 3 \ - \ / \ 3 \ *$$

.....

.....

.....

..... [3]

(b) Write this infix expression in RPN:

$$((7 + 3) - (2 * 8)) / 6$$

.....

.....

.....

..... [2]

(c) Evaluate this RPN expression:

$$a \ b \ - \ c \ d \ + \ * \ e \ /$$

when

$$a = 17, \ b = 5, \ c = 7, \ d = 3 \text{ and } e = 10$$

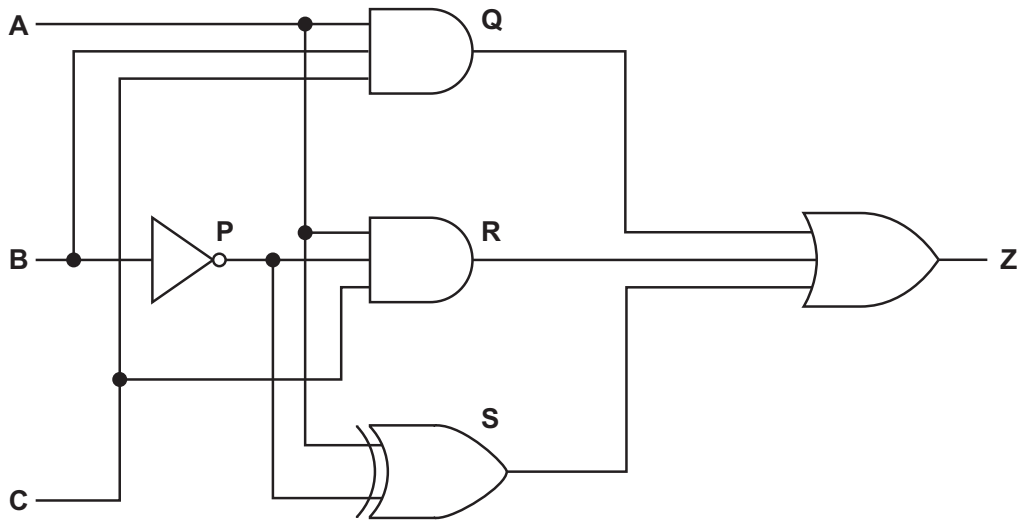
Show the changing contents of the stack as the RPN expression is evaluated.

[4]





6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.

Show your working.

			Working space				
A	B	C	P	Q	R	S	Z
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) Write the Boolean expression that corresponds to the logic circuit as a sum-of-products.

Z =

 [2]





(c) (i) Complete the Karnaugh map (K-map) for the Boolean expression:

$$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$$

BC		00	01	11	10
A	0				
	1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part (c)(ii) as a simplified sum-of-products.

.....
 [1]

7 (a) Describe what is meant by a digital certificate.

.....

 [3]

(b) Explain the role of a digital certificate in creating a digital signature.

.....

 [2]





- 8 A **declarative** programming language is used to represent the features that are available and the features that are unavailable on different body styles of a car.

```

01 feature(sunroof).
02 feature(automatic_tailgate).
03 feature(heated_seats).
04 feature(extra_seats).
05 feature(reversing_camera).
06 feature(dashboard_camera).
07 feature(air_conditioning).
08 feature(heated_windscreen).
09 feature(satnav).
10 bodystyle(saloon).
11 bodystyle(hatchback).
12 bodystyle(estate).
13 bodystyle(minivan).
14 bodystyle(convertible).
15 available(sunroof, hatchback).
16 available(sunroof, minivan).
17 available(reversing_camera, hatchback).
18 available(extra_seats, minivan).
19 available(reversing_camera, saloon).
20 unavailable(sunroof, convertible).
21 unavailable(automatic_tailgate, saloon).
22 unavailable(extra_seats, hatchback).

```

These clauses have the meanings:

Clause	Meaning
01	Sunroof is a feature.
10	Saloon is a body style.
15	Sunroof is available on a hatchback.
20	Sunroof is unavailable on a convertible.

- (a) Sliding doors is a feature that is available on a minivan but unavailable on a hatchback.

Write additional clauses to represent this information.

23

24

25

[3]



DO NOT WRITE IN THIS MARGIN



(b) Using the variable `Options`, the goal:

```
available(Options, saloon)
```

returns

```
Options = reversing_camera
```

Write the result returned by the goal:

```
available(Options, hatchback)
```

```
Options = ..... [1]
```

(c) `F` may be available for `B` if `F` is a feature and `B` is a body style and `F` is **not** unavailable for that body style.

Write this as a rule:

```
may_choose_option(F, B)
```

```
IF .....  
.....  
.....  
..... [4]
```

9 Explain what is meant by **Deep Learning** in relation to Artificial Intelligence (AI).

```
.....  
.....  
.....  
.....  
.....  
..... [3]
```





- 10 (a) State a condition that must be true for an array to be searchable for a binary search.

.....
 [1]

- (b) Complete the given pseudocode to find an item in a 1D array `Names` of type `STRING` using a binary search.

```

DECLARE Names : ARRAY[1:100000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE CurrentItem : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN
TopOfList ← 1
EndOfList ← 100000

OUTPUT "Which name do you wish to find? "
INPUT ToFind

.....

NotInList ← FALSE

WHILE ..... AND .....
    CurrentItem ← (TopOfList + EndOfList) DIV 2

    IF ..... THEN
        Found ← TRUE
    ELSE
        IF TopOfList >= EndOfList THEN

            .....

        ELSE
            IF ToFind > Names[CurrentItem] THEN

                .....

            ELSE
                EndOfList ← CurrentItem - 1
            ENDIF
        ENDIF
    ENDIF
ENDWHILE
IF Found = TRUE THEN
    OUTPUT "Item found at position ", CurrentItem, " in array"
ELSE
    OUTPUT "Item not in array"
ENDIF
  
```

[5]





- (c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

.....

.....

.....

..... [2]

- 11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

- (a) State **two** features of RISC processors.

.....

.....

.....

..... [2]

- (b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

.....

.....

.....

.....

.....

..... [3]

- (c) Explain how pipelining affects interrupt handling for RISC processors.

.....

.....

.....

.....

.....

..... [3]





Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

